

Complex oscillatory system

Y13 (2013) — English translation

The mass m_1 is connected by threads to two identical masses m_2 . The threads are thrown through small light blocks, the distance between which is L .

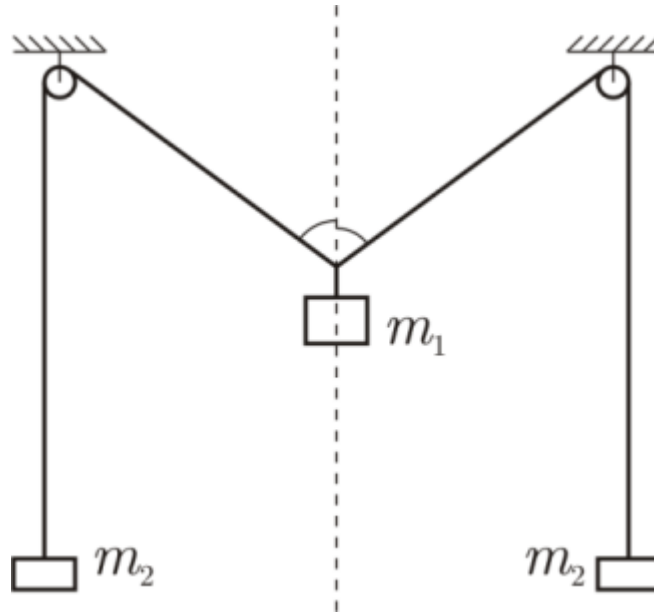


Fig. 1.

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| A1 | At what mass ratio m_1/m_2 will the system be in a stable equilibrium position (mass loads m_2 do not touch the blocks)? | 0 |
| A2 | Find the period of small oscillations of such a system (the mass m_1 oscillates along a vertical straight line). | 0 |
| A3 | Consider the special case of oscillations when $m_1 = m_2$. | 0 |

Source: <https://pho.rs/p/3995>